

Objective function

Convexity

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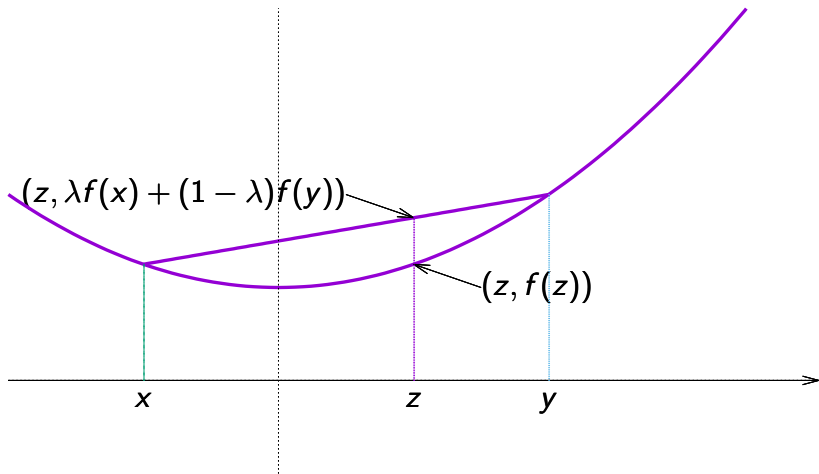
Optimization: principles and algorithms

Convexity

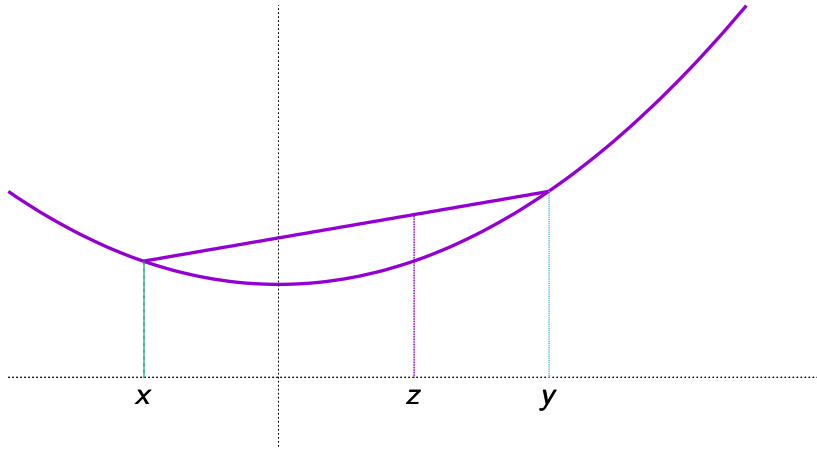
Before solving the problem, let's investigate its geometry. We start by analyzing the objective function f . The first concept that is important for optimization is convexity.

Convexity

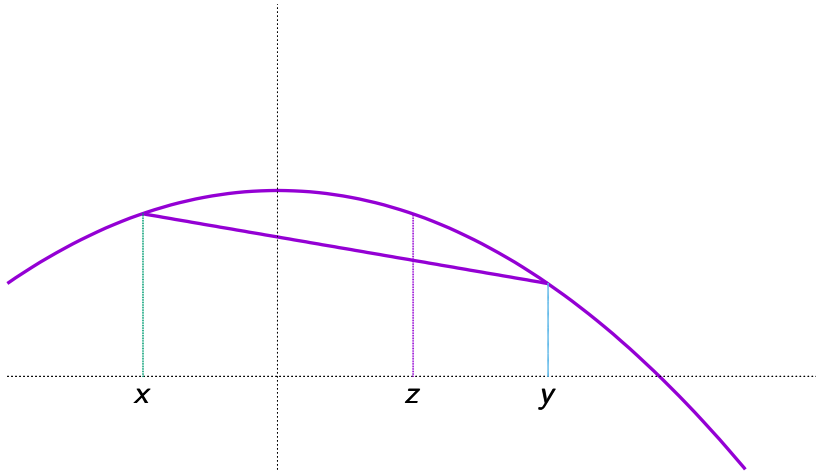
$$z = \lambda x + (1 - \lambda)y, \quad 0 \leq \lambda \leq 1$$



Convexity



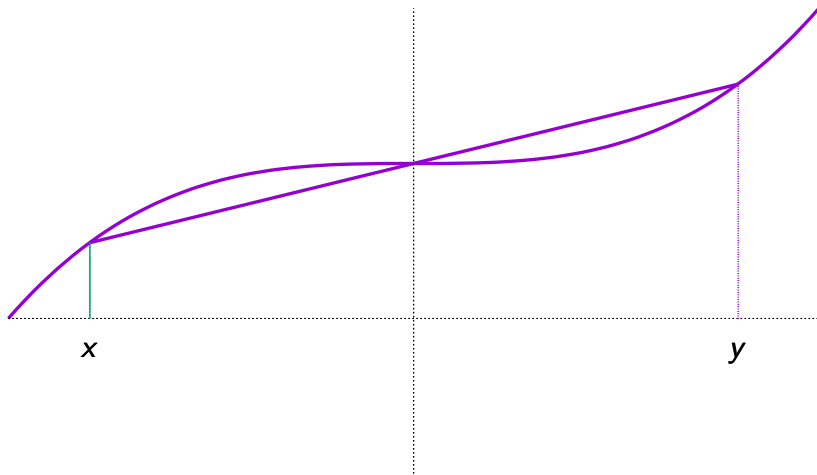
Concavity



Convexity and concavity

Mention that concavity is not the same as “non convexity”

Convexity and concavity



Summary

Convexity is an important property of the objective function.

In particular, when the objective function is convex, it is possible to find a global optimum.

Therefore, it is advised to model the problem as a convex problem whenever possible.